Free-body diagram tic-tac-toe

15 D

DIRECTIONS · Complete three in a row. Draw every diagram as a dot with labeled arrows.

DRAW A book resting on a level table.	**DRAW** A lamp hanging at rest from one rope.	**EXPLAIN** Why is there no "force of motion" arrow on a rolling cart?
DRAW A box pulled right across a rough floor.	**CHALLENGE ★** A ball at the top of its toss (ignore air). Draw it. Explain.	**DRAW** A falling ball with air resistance.
EXPLAIN How do arrow lengths show balanced forces?	**REAL WORLD** Your phone sitting on your desk. Name each source.	**DRAW** A box sliding right after the push ends.

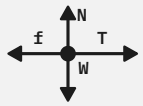
Three in a row: _____ · The square that taught me most: _____

S5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: free-body diagrams

Language goal: I can **label** every force on one object.



• with arrows

free-body diagram

diagrama de cuerpo libre · diagrama de corpo livre

only the forces ON one object

my language: _____

use it: _____



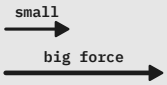
dot

punto · ponto

stands for the object

my language: _____

use it: _____



short → long

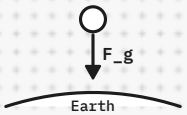
arrow length

largo de la flecha · comprimento da seta

shows force strength

my language: _____

use it: _____



↓ W

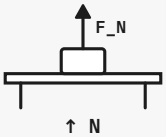
gravity W

gravedad · gravidade

Earth pulls down

my language: _____

use it: _____



↑ N

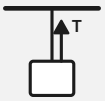
normal N

normal · normal

a surface pushes ⊥

my language: _____

use it: _____



↑ T

tension T

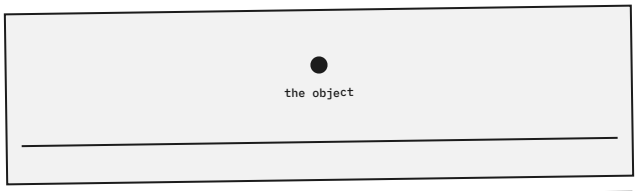
tensión · tração

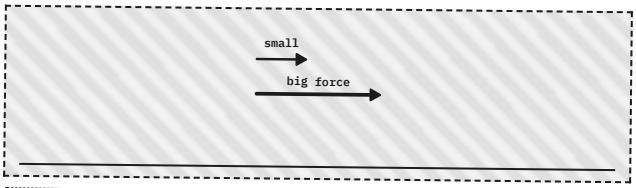
a rope pulls

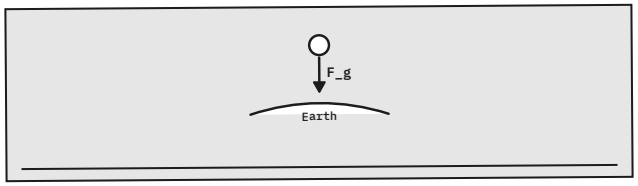
my language: _____

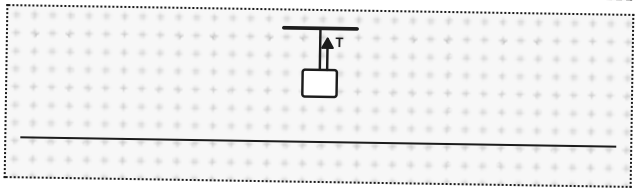
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner □

S5 (1) (2) (3)

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15 T

The phantom force

A student's FBD is on trial. Find the fake arrow.

EXHIBIT A · THE SCENE

A cart rolls right on a level, frictionless track after a push. The push has ended.

EXHIBIT B · THE STUDENT'S DIAGRAM

- ↑ normal 10 N
- ↓ weight 10 N
- → "push" 5 N
- → "motion" force

ANALYSIS 1 · CHECK SOURCES

For each arrow, name the object exerting it. Which arrows have no source?

ANALYSIS 2 · CORRECT FBD

THE SIX RULES · CHECK EACH

- ☐ dot for the object
- ☐ one arrow per force
- ☐ arrow points the force's way
- ☐ length ∝ strength
- ☐ label each arrow
- ☐ only forces ON it

THE VERDICT

The fake arrows are ____ because ____. The cart keeps rolling because ____

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Say it, then write it: FBDs

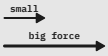
Language goal: I can **name** the source of each force.



free-body diagram
diagrama de cuerpo libre · diagrama de corpo livre
forces on one object



source
fuente · fonte
what exerts the force



arrow
flecha · seta
shows direction + size



on
sobre · sobre
acting on the object

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: object · source · arrow · length · motion

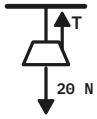
First I choose one ____. Every arrow needs a ____. The ____ points the way the force acts. Its ____ shows how strong it is. I never draw an arrow for ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The ____ exerts a ____ force on the ____.
The arrows are equal because the forces ____.

3 · SOLVE WITH THE STEPS



A lamp hangs at rest from one rope. It weighs 20 N.

STEP 1 · CHOOSE
object: ____

STEP 2 · LIST
sources: ____, ____

STEP 3 · DRAW + SAY
"Tension ____ N up, weight ____ N down."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ I drew ____ arrows because ____.
☐ There is no arrow for motion because ____.
☐ The arrows have the same length because ____.

I said it ☐ · I wrote it ☐

S5 1 2 3

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Four steps, every diagram

15 M

DIRECTIONS · Read, draw the FBD, solve, then write.

PROBLEM 1 · A 20 N lamp hangs at rest from one vertical rope. Draw its FBD and find the tension.

1 READ

object ____ · sources ____ · at rest → ____

2 DRAW

dot + arrows

3 SOLVE

$\Sigma F_y = 0$

4 WRITE

The tension is ____ N up, equal to ____ .

PROBLEM 2 · A box slides right after the hand lets go. Friction is 3 N. Weight and normal are 10 N. Draw the FBD and find the net force.

1 READ

forces ____ · direction of motion ____

2 DRAW

dot + arrows

3 SOLVE

net = sum with signs

4 WRITE

The net force is ____ N ____, so the box ____ .

Hardest box: □ Read □ Draw □ Solve □ Write

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Does it belong on the FBD?

15 N

DIRECTIONS · Object: a box pulled right by a rope across a rough floor. Sort each card.

A tension from the rope	B the box's velocity	C weight from Earth	D the box pushes the floor
E normal force from the floor	F friction from the floor	G "the force of motion"	H the hand pulling the rope
I air resistance (if fast)	J the box's mass	K the rope pulls the hand	L acceleration

BELONGS	DOESN'T BELONG	DEPENDS
a force ON the box	not a force, or not on the box	explain below

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

S5 1 2 3

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